

Experiment 4: Launching a Flask Web Application Locally

Aim

To create a Python virtual environment, install the Flask dependencies for the App Engine application, and launch the web application locally.

Tools Required

- Ubuntu VM
- Python 3
- Project created in Experiment 3
- python3-venv
- Flask 3.1.3

Detailed Procedure

Open Terminal and move to the gae-helloworld directory.

Update the package repository.

Install the Python virtual-environment packages.

```
cd ~/gae-helloworld  
sudo apt update  
sudo apt install python3-venv python3-full -y
```

Create the virtual environment.

```
python3 -m venv venv
```

Activate the environment.

```
source venv/bin/activate
```

Confirm that the terminal prompt indicates the virtual environment is active.

Install the project dependencies.

```
pip install -r requirements.txt
```

Verify the installed Flask and Werkzeug versions.

```
pip show flask  
pip show werkzeug
```

Run the application.

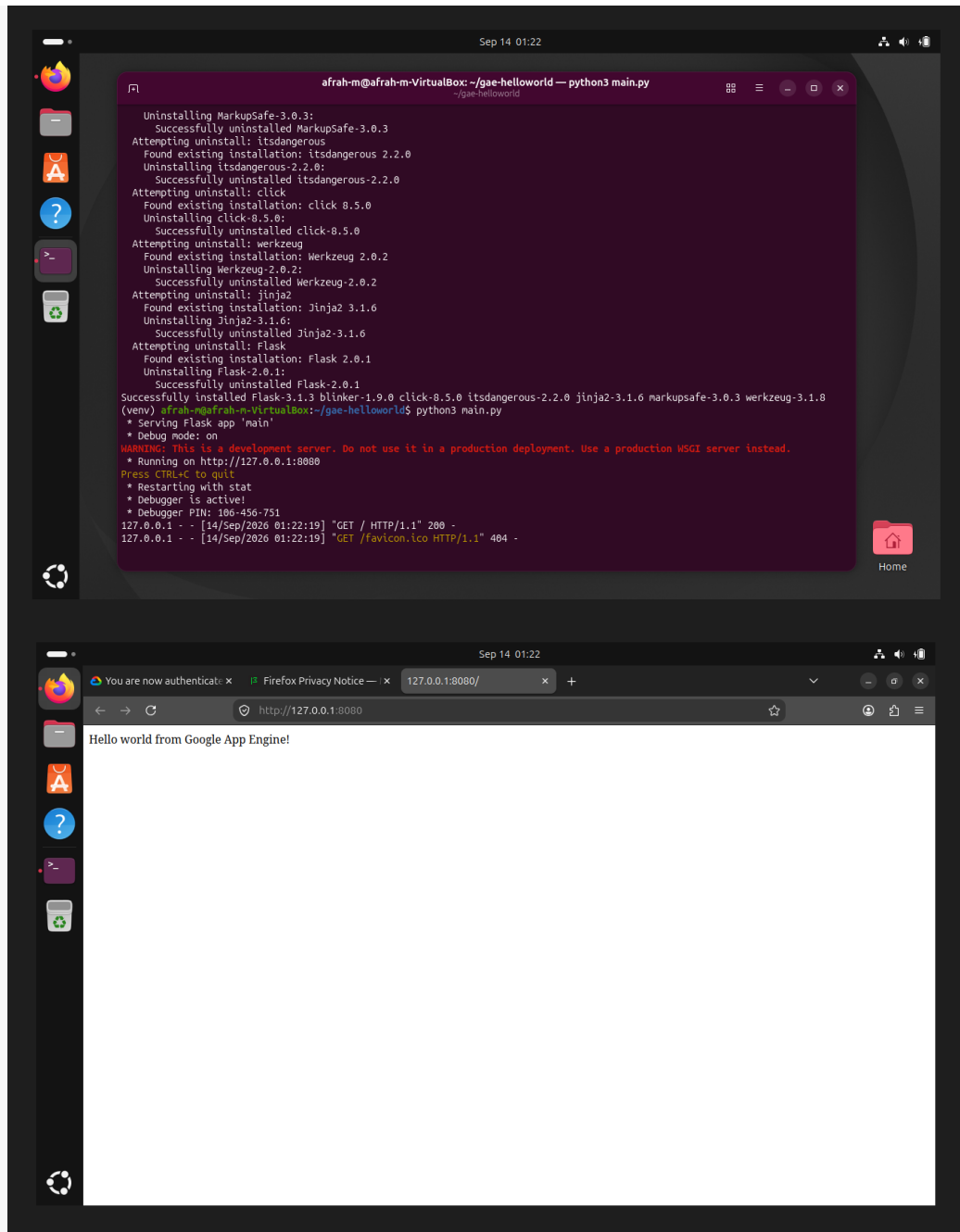
```
python3 main.py
```

Open a browser inside the Ubuntu VM and visit <http://127.0.0.1:8080>. Verify that the page displays the Hello World message.

Important Compatibility Note

The completed workflow used Flask 3.1.3 with a compatible Werkzeug release. The older Flask 2.0.1 dependency from the original manual produced compatibility errors, so it was replaced by Flask 3.1.3.

Output



The screenshot displays two windows from a virtual machine. The top window is a terminal titled 'afrah-m@afrah-m-VirtualBox: ~/gae-helloworld -- python3 main.py'. It shows the process of uninstalling several Python packages (MarkupSafe, itsdangerous, click, Werkzeug, Jinja2, Flask) and then installing Flask, blinker, click, itsdangerous, Jinja2, markupsafe, and Werkzeug. After successful installation, it runs 'python3 main.py', which serves the Flask app 'main' on http://127.0.0.1:8080. The terminal output includes a warning about using a development server and shows log entries for GET requests to '/' and '/favicon.ico'. The bottom window is a web browser with the address bar showing 'http://127.0.0.1:8080/'. The browser page displays 'Hello world from Google App Engine!'.

```
afrah-m@afrah-m-VirtualBox: ~/gae-helloworld -- python3 main.py
Uninstalling MarkupSafe-3.0.3:
Successfully uninstalled MarkupSafe-3.0.3
Attempting uninstall: itsdangerous
Found existing installation: itsdangerous 2.2.0
Uninstalling itsdangerous-2.2.0:
Successfully uninstalled itsdangerous-2.2.0
Attempting uninstall: click
Found existing installation: click 8.5.0
Uninstalling click-8.5.0:
Successfully uninstalled click-8.5.0
Attempting uninstall: werkzeug
Found existing installation: Werkzeug 2.0.2
Uninstalling Werkzeug-2.0.2:
Successfully uninstalled Werkzeug-2.0.2
Attempting uninstall: Jinja2
Found existing installation: Jinja2 3.1.6
Uninstalling Jinja2-3.1.6:
Successfully uninstalled Jinja2-3.1.6
Attempting uninstall: Flask
Found existing installation: Flask 2.0.1
Uninstalling Flask-2.0.1:
Successfully uninstalled Flask-2.0.1
Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.5.0 itsdangerous-2.2.0 Jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8
(venv) afrah-m@afrah-m-VirtualBox:~/gae-helloworld$ python3 main.py
* Serving Flask app 'main'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8080
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 106-456-751
127.0.0.1 - - [14/Sep/2026 01:22:19] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Sep/2026 01:22:19] "GET /favicon.ico HTTP/1.1" 404 -
```

127.0.0.1:8080/

http://127.0.0.1:8080

Hello world from Google App Engine!

Result

The Flask web application was launched successfully in the Ubuntu VM and the Hello World page was displayed in the browser.